데이터분석기법

13주차: Convolutional neural networks and represenation learning

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Goals for Today

- Understand the limitations of fully connected neural networks in computer vision
- Learn the core principles behind convolutional neural networks, including local connectivity and parameter sharing
- Explore the building blocks of CNNs: convolution, activation, pooling, normalization, and dropout
- Introduce representative CNN architectures
- Familiarize with key computer vision tasks

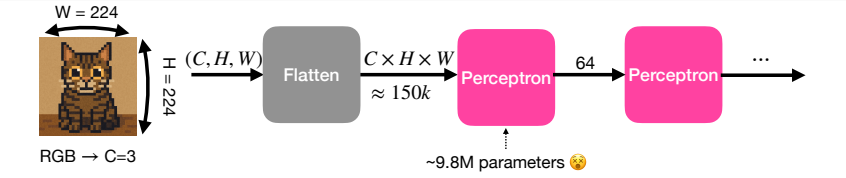
Computer Vision

Computer Vision

- Computer vision (CV) is a field of AI that enables machines to interpret and understand visual information from the world: 🤖 + 👀
- Traditional CV methods rely on hand-crafted features: Scale-invariant feature transform (SIFT), Histogram of oriented gradients (HOG), and etc.
- These features often failed under real-world variability

A pedestrian image (left) and its gradient (right). Source: [11]

DL for CV?



- Loss of spatial structure
  - 2D images are 3D arrays, but fully connected neural networks (FCNNs) require 1D input
  - Flattening destroys the spatial relationships between pixels
  - FCNNs treat every pixel independently, ignoring local correlations like edges
  - If a pattern (like an edge) shifts slightly in position, the FCNN sees it as a completely different input
- Parameter explosion
  - For high-resolution images, the number of parameters becomes massive
  - A single perceptron layer applied to a 3×224×224 image requires about 9.8 million parameters to produce a 64-dim hidden vector

Convolutional Neural Networks

Core Ideas

- **Convolutional Neural Networks (CNNs)** are a type of neural network designed to process data with a grid-like structure, such as 1D time-series or 2D images
- CNNs are particularly effective at capturing local patterns using **convolution** operations[1]
- **Local Connectivity**: convolution extracts local features from small subregions of the image
- **Parameter sharing**: convolution uses shared parameters to detect local features throughout the entire image

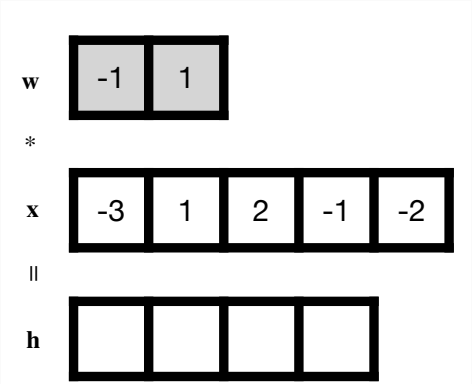
[1] Convolutional neural networks typically use *cross-correlation* instead of convolution, but since the kernels are learnable and the distinction is minor, we refer to cross-correlation as "convolution" throughout this lecture for simplicity.

1D Discrete Convolution

Convolution

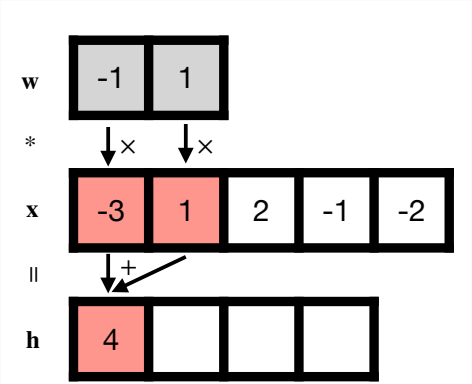
- **Input**, $\mathbf{x} \in \mathbb{R}^M$ (e.g. $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5]$)
- **Kernel** or **filter**, $\mathbf{w} \in \mathbb{R}^N$ (e.g. $\mathbf{w} = [w_1, w_2]$)
- **Output (feature map)**, $\mathbf{w} * \mathbf{x}$ or $(\mathbf{w} * \mathbf{x})_i = \sum_{n=1}^{N} w_n x_{i+n}$
  - e.g. $\mathbf{w} * \mathbf{x} = (w_0 x_0 + w_1 x_1, w_0 x_1 + w_1 x_2, w_0 x_2 + w_1 x_3, w_0 x_3 + w_1 x_4)$

Visualization of 1D Discrete Convolution



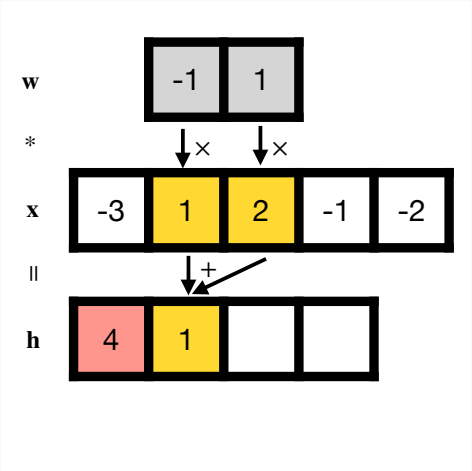
- Kernel: **w** / Input: **x** / Output feature map: **h**

Visualization of 1D Discrete Convolution (cont.)



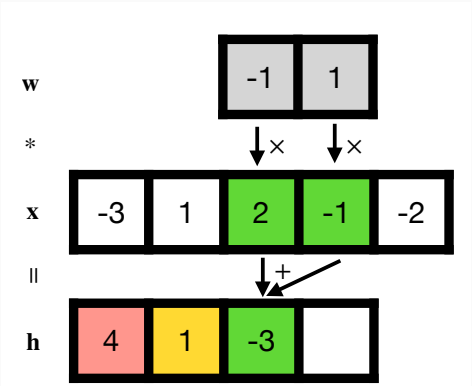
- Kernel: **w** / Input: **x** / Output feature map: **h**

Visualization of 1D Discrete Convolution (cont.)



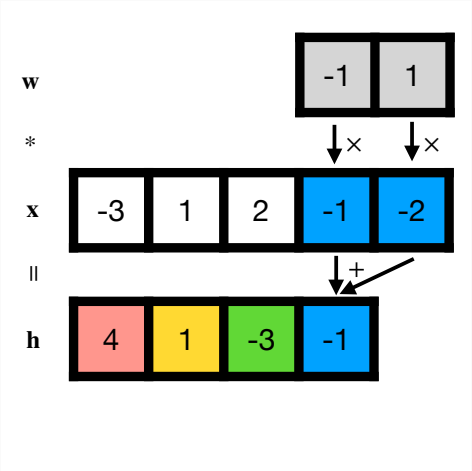
- Kernel: **w** / Input: **x** / Output feature map: **h**

Visualization of 1D Discrete Convolution (cont.)



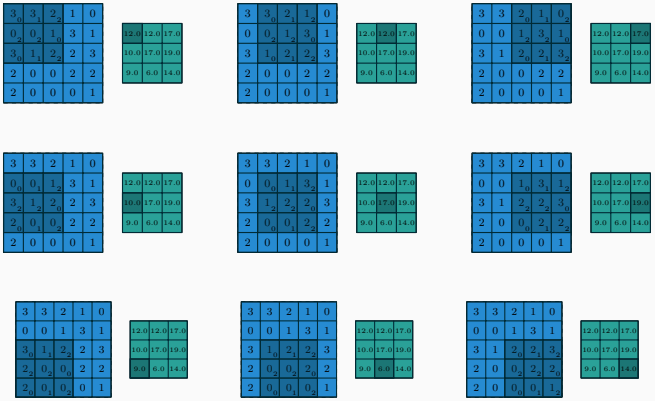
- Kernel: **w** / Input: **x** / Output feature map: **h**

1D Discrete Convolution (cont.)



- Kernel: **w** / Input: **x** / Output feature map: **h**

2D Discrete Convolution (cont.)

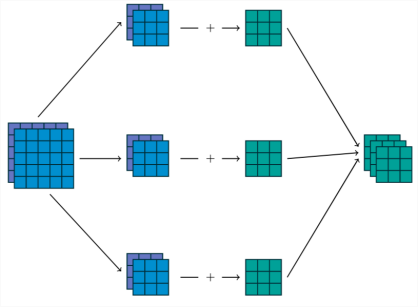


Source: [1]

- 2D discrete convolution: $(W * X)_{ij} = \sum_h \sum_w W_{h,w} X_{i+h,j+w}$

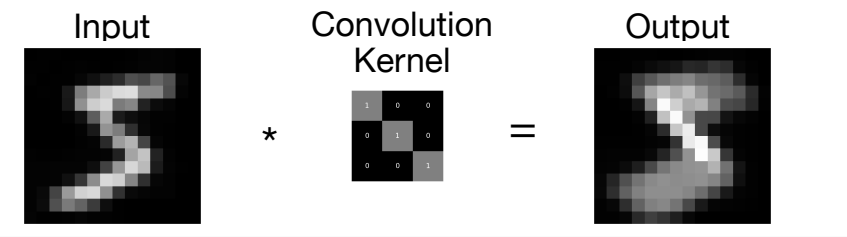
Multi-Channel Convolution

- Input images can have multiple channels, such as RGBA images with four channels: red, green, blue, and alpha (transparency)
- A convolution takes a multi-channel input (leftmost stack of purple-blue squares), and applies multiple kernels (small purple-blue squares)
- In each path, each kernel processes the corresponding input channel, and the results are summed element-wise to produce one channel of the output feature map
- Combining multiple such outputs results in a multi-channel output (rightmost stack of green squares)
- This enables the construction of deeper and more expressive models



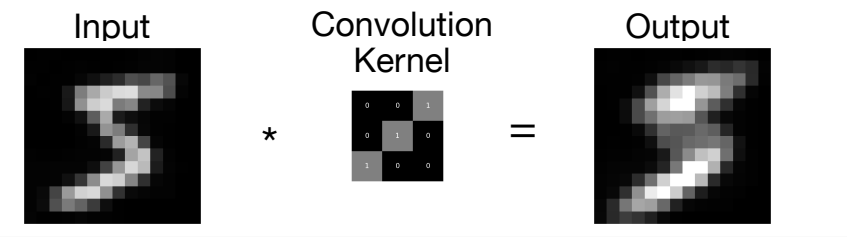
Source: [1]

Convolution as Edge Detection



- **Input** is a grayscale image of the digit 5, represented as a 2D array of pixel intensities, where brighter pixels indicate higher intensity, and darker pixels indicate lower intensity
- **Convolution kernel** is a small 3×3 matrix that slides over the input image to detect edges
- **Output feature map** is the result of applying the kernel to the input image, highlighting edges where intensity changes align with the kernel's pattern

Convolution as Edge Detection (cont.)



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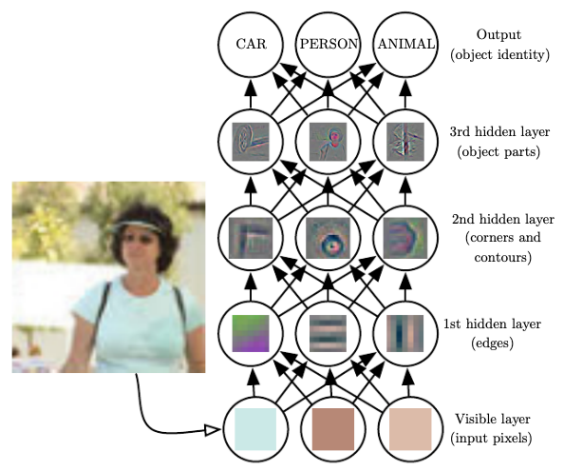
Convolution as Edge Detection (cont.)

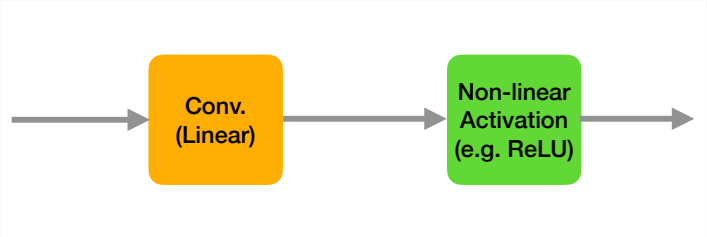


Source: [13, 3]

Convolution Arithmetic

To gain a deeper understanding of convolution, refer to the following link:
https://github.com/vdumoulin/conv_arithmetic

Linearity of Convolution

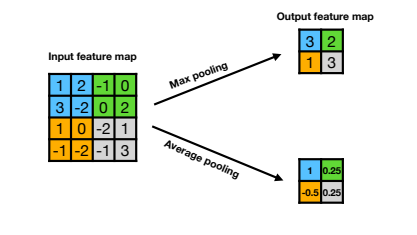


- Convolution is a linear operation

$$\mathbf{w} * (a\mathbf{u} + b\mathbf{v}) = a\mathbf{w} * \mathbf{u} + b\mathbf{w} * \mathbf{v} \tag{1}$$

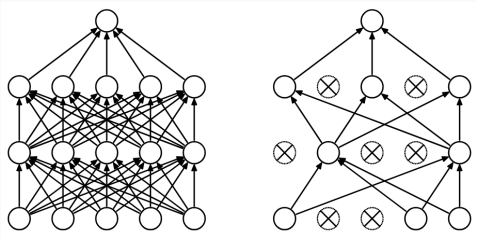
- In convolutional neural networks (CNNs), a convolution layer is typically followed by a non-linear activation function, just like in fully connected neural networks

Pooling



- Pooling is a downsampling operation that reduces the spatial dimensions of feature maps, making the model more translation invariant and reduces computational cost
- A pooling window (e.g., 2×2) slides across the input feature map
- For each window, it applies an aggregation function, such as max and average
- PyTorch provides a bunch of pooling methods (link)

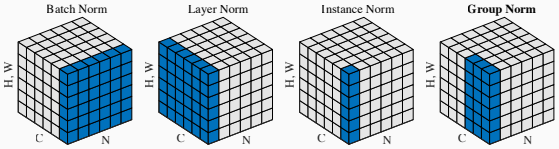
Dropout



Left: Standard Neural Net. Right: After applying dropout. Source: [9]

- Dropout is a regularization technique where, during training, a random subset of neurons is ”dropped” (set to zero) in each layer, preventing overfitting by encouraging the network to learn redundant representations
- Undropped nodes are scaled up by $\frac{1}{1-p_{\text{dropout}}}$ during training to ensure the network's output remains consistent in scale
- See torch.nn.Dropout

Normalization



Source: [12]

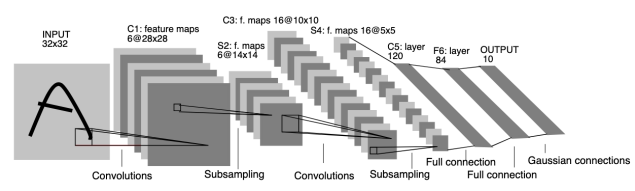
- **Normalization** is a technique used to improve the training of DL models by adjusting the inputs of each layer to have a mean of zero and a standard deviation of one, thereby stabilizing the learning process
- During training, a normalization layer uses the mean and standard deviation of each batch for backpropagation
- In the inference phase, a normalization layer uses a running average of the means and standard deviation from the training phase, maintaining stable outputs
- Batch normalization is widely used in DL models for image data
- See Normalization Layers in PyTorch

CNN Demo

ConvNetJS CIFAR-10 demo

Architectures

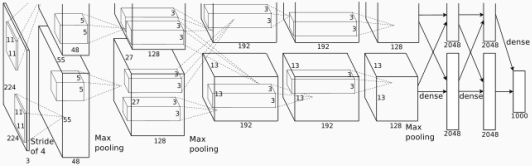
LeNet



Architecture of LeNe-5 [7]

- Paper: LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (2002). Gradient-based learning applied to document recognition. Proceedings of the IEEE, 86(11), 2278-2324.
- Task: Digit recognition (MNIST)
- Architecture: Alternating convolution and pooling layers followed by fully connected layers
- Key Idea: Introduced the basic CNN structure—convolution, pooling, and fully connected (FC) layers for image classification

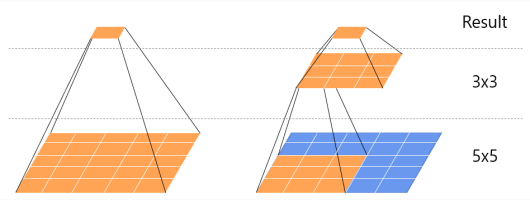
AlexNet



Architecture of AlexNet [6]

- Paper: Krizhevsky, A., Sutskever, I., Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. Advances in neural information processing systems, 25.
- Task: Image classification
- Architecture: 5 conv. + 3 FC layers, ReLU activation, dropout, and overlapping max pooling
- Key Idea: Demonstrated that deep CNNs can significantly outperform traditional methods on large datasets using GPU acceleration

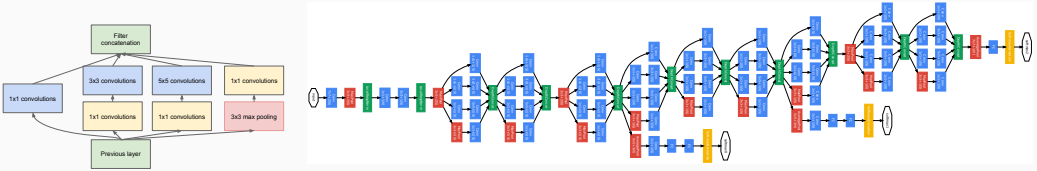
VGGNet



A comparison of the 5×5 and 3×3 convolution filters

- Paper: Simonyan, K., Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556.
- Task: Image classification
- Architecture: Very deep network (up to 19 layers) using only 3×3 convolutions and 2×2 max pooling
  - one 5×5 conv.: $5 \times 5 = 25$ parameters / two 3×3 conv.: $2 \times 3 \times 3 = 18$ parameters
- Key Idea: Showed that depth and simplicity (stacking small filters) can improve performance

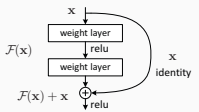
GoogLeNet

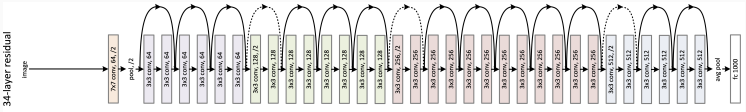


Source: [10]

- Paper: Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., et al. (2015). Going deeper with convolutions. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 1-9).
- Image classification
- Architecture: Introduced the Inception module, combining multiple filter sizes (1×1, 3×3, 5×5) and max pooling in parallel
- Key Idea: Efficiently captures multi-scale features while reducing parameters with 1×1 convolutions

ResNet

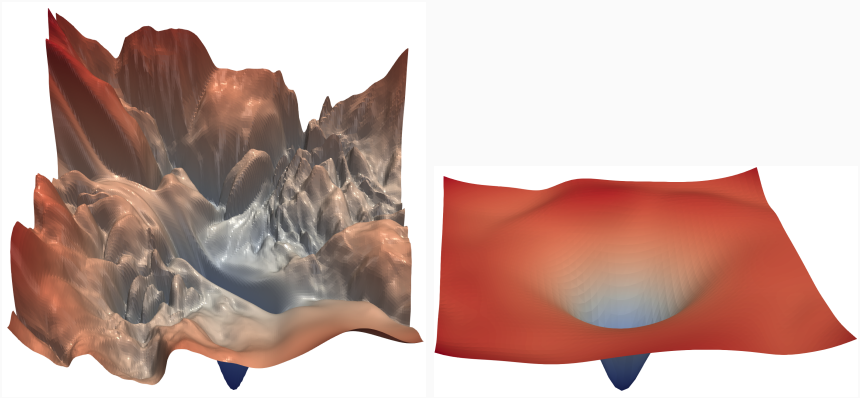




Source: [5]

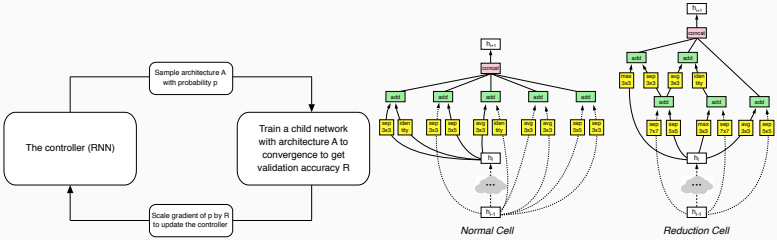
- Paper: He, K., Zhang, X., Ren, S., Sun, J. (2016). Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 770-778).
- Task: Image Classification
- Key Idea: Enables training of very deep networks by addressing the vanishing gradient problem with residual connections
- Architecture: Very deep network (up to 152 layers) using residual connections

ResNet (cont.)



The loss surfaces of ResNet-56 (left) without and (right) with residual connection. Source: [8]

NASNet



Source: [14]

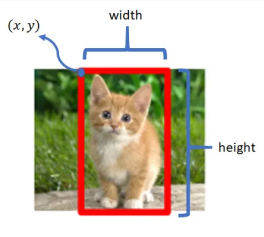
- Paper: Zoph, B., Vasudevan, V., Shlens, J., Le, Q. V. (2018). Learning transferable architectures for scalable image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 8697-8710).
- Task: Image classification
- Key Idea: Use Neural Architecture Search (NAS) with reinforcement learning to find the best convolutional cell structures

CNN Model Zoo

- Pretrained DL models for CV by PyTorch Teams

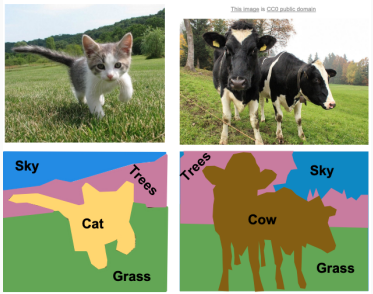
CV Tasks

Object Detection



- Goal: Both detect the presence of objects and predict their positions
  - $\rightarrow$ Classification + Regressing bounding box coordinates
- Example: Detect a cat in an image

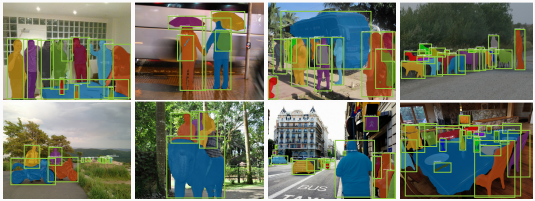
Semantic Segmentation



Source: [2]

- Goal: Assign a class label to each pixel in the image
  → pixel-wise classification
- Example: Identify which pixels belong to roads, pedestrians, cars, etc., in a self-driving car scenario
- Models: FCN, U-Net, DeepLab

Instance Segmentation



Source: [4]

- Segment each individual object instance, even if they are of the same class
  - → semantic segmentation + object detection
- Segment two different cats in the same image
- Popular Models: Mask R-CNN

Keypoint Detection

Source: [4]

- Goal: Detect specific points (e.g., joints) on objects or human bodies
- Example: Detect human joints like knees, elbows, etc.
- Popular Models: Mask R-CNN, OpenPose, HRNet

Summary

- CNNs address the limitations of FCNNs by preserving spatial structure and reducing parameters via local connectivity and weight sharing
- Convolution layers extract local features, often visualized as edge detectors
- CNNs power key CV tasks like classification, object detection, segmentation, and keypoint detection

Questions**?**

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